When constant g is not enough

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Recognize when mgh works near a planet's surface and when it stops working well.

SUCCESS CRITERIA - I can...

1. Compare altitude with planetary radius.
2. Explain that gravitational field weakens with center distance.
3. Choose an equation that fits the height change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "assumption" and "field" to explain your answer. Sentence starters are optional.

assumption: a condition accepted for using a model

field: a description of an influence at each location

potential energy: energy associated with the configuration of interacting objects

01 Warm-up: bring back an earlier idea

A **3 kg** cart starts at rest. A **19.5 N** pull and **7 N** resistance act over **3 m**. **Find final speed**.

Word help: final = at the end of the interval; subscript f

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

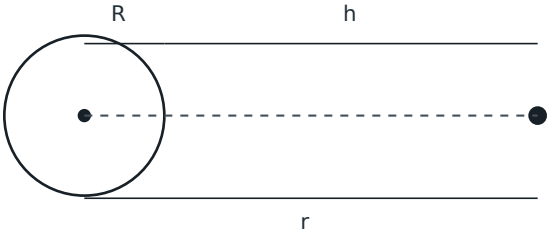
02 Worked example: explain the prediction

Reference: $\Delta U \approx mg\,\Delta h$; $g = GM/r^2$.
The first relation needs nearly constant g. For large changes in r, the field changes; one fixed surface value is insufficient.

Worked reasoning: A calculation near the surface treats g as constant over small height changes. At a center distance twice a planet's radius, however, g is only one quarter of its surface value.

Your turn: 1. **Sketch a planet** and **mark center distances R** and **2** R.
2. **Explain why one surface g** cannot describe the entire trip exactly.

Sentence starter: My design includes ... . It does what is needed by ... .

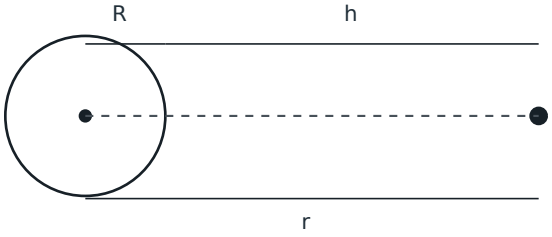


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Compare lifting a book 1 m** with moving a satellite one planetary radius above the surface.
2. Which better fits constant g and why?

Sentence starter: Both ... . They differ in ... because ... .

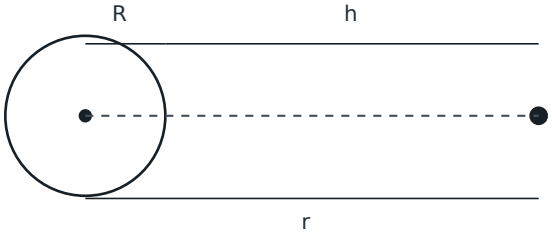


Marzano check - circle: ① ② ③

04 Apply the physics

A gravity simulation uses GM = **100 m³/s²**. A **2 kg** object moves from center distance **10 m** to **20 m**.
1. **Label both distances**.
2. Calculate $\Delta U = GMm(\mathbf{1}/r_i - \mathbf{1}/r_f)$.
3. Compare with mgh using the starting field g = **1 m/s²** and h = **10 m**.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is altitude h not interchangeable with center-to-center r in the universal expression?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

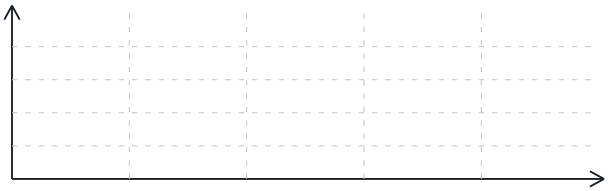
06 Review: connect an earlier idea

Total momentum initially is (**24**, **0**) kg·m/s. One fragment leaves with (**4**, **6**) kg·m/s. **Find the other momentum vector** and magnitude.
Label momentum components and their sum with units.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

North momentum



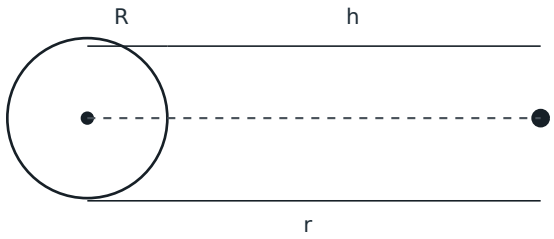
East momentum

Marzano check - circle: 1 2 3

07 Exit check

Using the same simulated planet, compare gravity at r=**10** and **20 m**. Connect the result to the choice of energy equation. **Label the relevant heights or** distances on the diagram.

Sentence starter: Both … . They differ in … because … .

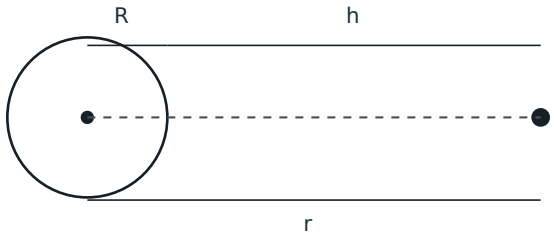


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A planet has GM=**100 m³/s²** and radius **5 m**.
1. **Compare exact gravitational potential-energy increase** for a **2 kg** object raised to r=**10 m** with the surface-g approximation.
2. **Label the relevant heights or** distances on the diagram.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3